STROKE PREDICTION USING LOGISTIC REGRESSION

Project submitted to

St Thomas College Palai (Autonomous)

by

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Reg. No. 24PSTAT0033

In partial fulfilment of the requirements for the degree of

MASTER OF SCIENCE

in

STATISTICS

awarded by

Mahatma Gandhi University, Kottayam

Under the supervision and guidance of

Prof. SEEMON THOMAS

Professor

Department of Statistics

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March 2026

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CERTIFICATE

This is to certify that the project entitled "**Stroke Prediction Using Logistic Regression**" is a Bonafide record of research work carried out by Ms**. Bhavya Prakasam (Reg. No. 24PSTAT0033)**, under my supervision and guidance, in partial fulfilments of the requirements for the award of degree of **Master of Science in Statistics** at St Thomas College Palai Autonomous.

It is further certified that this work has not previously formed the basis for the award of any degree, diploma, fellowship, or any other similar title of any university or institution.

She is hereby permitted to submit the project.

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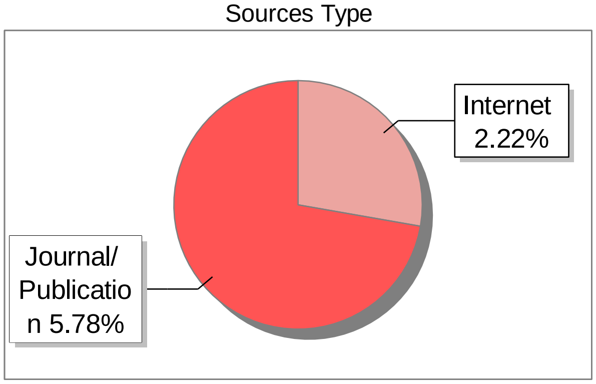

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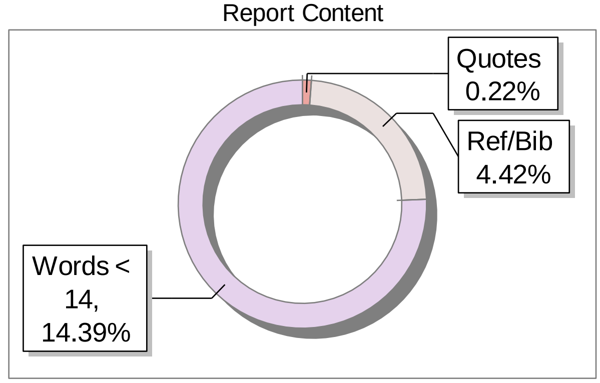

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Palai **Bhavya Prakasam**

March 2026

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I would like to express my sincere gratitude to everyone who has supported and guided me in the successful completion of this project titled "Stroke Prediction using Logistic Regression."

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Bhavya Prakasam

Palai
March 2026

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

Stroke is a serious medical condition that occurs when the blood supply to a part of the brain is reduced, preventing brain tissue from receiving the oxygen and nutrients it needs. Within minutes, brain cells start to die, and because of this, strokes are considered medical emergencies. Stroke is the 2nd leading cause of death globally, responsible for approximately 11% of total deaths. Strokes have a massive effect on a person's quality of life, and they are a great burden on the systems of health care worldwide.

There are mainly two types of stroke: i) Ischemic stroke, there is a blockage in a blood vessel in the brain, and ii) Hemorrhagic stroke, which occurs due to the rupture of a blood vessel. In addition to these, a transient ischemic attack (TIA), often referred to as a “mini-stroke”, which is considered to be a precursor to a stroke. Because of this, the early detection of a TIA is extremely important in the prevention of a potential stroke. Therefore, early detection and prevention play a crucial role in reducing the risk and severity of this condition.

Risk factors for stroke can be divided into two categories; non-modifiable and modifiable. Non-modifiable factors include age, gender, and genetic predisposition; while modifiable factors include high blood pressure, diabetes, heart disease, smoking, obesity, and high cholesterol. Among these, age is considered one of the most significant factors, as the risk of stroke increases with increase in age.

With the increasing availability of healthcare data, statistical and machine learning techniques are being widely used to predict the likelihood of stroke. In this context, logistic regression is one of the most commonly used methods due to its effectiveness in handling binary classification problems.

Logistic regression is used to create a regression model when the dependent variable(target) is categorical. In stroke prediction problem statement, dependent variable is binary ‘0’ denoting absence of stroke and ‘1’ denoting presence of stroke is fitted into a regression model to estimate the probability of stroke. The coefficient of regression are estimated in such a way that predicted probability always lies between 0 to 1 using logistic function. Logistic regression has an interpretable model that explains how different risk factors are contributing towards stroke.

Objective of the current study is to identify relationship between different risk factors and predict whether a person is prone to stroke or not using logistic regression analysis. All the factors like age, gender, BMI, glucose level etc are statistically analyzed to identify most important factors on which stroke occurs. Results of this analysis can help in early diagnosis and better risk assessment that can help implement preventive measures that improve healthcare.

1.2 OBJECTIVES OF THE STUDY

The main objectives of the study are:

• To study the stroke dataset and find out the distribution of various attributes like age, gender, body mass index, glucose level etc.

• To identify and study the main risk factors of stroke.

• Build a predictive model by using logistic regression to predict the probability of stroke.

• Evaluate the logistic regression model using appropriate evaluation metrics such as accuracy, confusion matrix etc.

•To interpret the clinical significance of the model results by identifying the most influential risk factors through the analysis of Odds Ratios.

1.3 LITERATURE REVIEW

Stroke has been widely recognized as a major global health concern due to its high mortality and long-term disability rates. According to the World Health Organization, stroke is a cerebrovascular condition caused by an interruption in blood supply to the brain, and its occurrence is strongly influenced by both medical and lifestyle-related risk factors.

Global studies have consistently highlighted the increasing burden of stroke across populations. The work of Valery L. Feigin and colleagues (2014) demonstrated that stroke remains one of the leading causes of death and disability worldwide, emphasizing the need for early detection and prevention strategies. Similarly, the INTERSTROKE study conducted by Martin J. O'Donnell et al. (2010) identified key modifiable risk factors such as hypertension, smoking, and diabetes as major contributors to stroke risk across different countries.

Early epidemiological research, particularly the Framingham Heart Study, has played a crucial role in understanding stroke risk factors. Studies by William B. Kannel and colleagues (1994), as well as Philip A. Wolf et al. (1991), highlighted the strong association between hypertension, atrial fibrillation, and stroke. These findings established that cardiovascular conditions significantly increase the likelihood of stroke occurrence.

From a methodological perspective, logistic regression has been widely used as an effective statistical tool for predicting binary outcomes such as stroke. The book Applied Logistic Regression by David W. Hosmer and Stanley Lemeshow (2000) provides a strong theoretical foundation for applying logistic regression in medical research.

Empirical studies have also applied logistic regression in stroke prediction. For instance, Leila Amini et al. (2019) developed a logistic regression model to predict stroke in an Iranian population, demonstrating that factors such as age, hypertension, and glucose levels significantly influence stroke risk.

In addition to statistical modeling, modern data analysis tools and software have played an important role in predictive studies. Packages such as caret, introduced by Max Kuhn (2008), support model building and validation, while visualization tools like ggplot2 developed by Hadley Wickham (2016) enhance data interpretation. Furthermore, the pROC package by Xavier Robin et al. (2011) facilitates the evaluation of model performance using ROC curves and AUC.

1.4 STATISTICAL TOOLS USED

1. Descriptive Statistics

Descriptive statistics were used to analyze and present fundamental features of the data set. Statistical measures like mean, median, minimum, maximum, and percentages were calculated to gain information about variables distribution.

2. Data Visualization

Graphical approaches like histogram, bar chart box plot etc were employed for data visualization. This approach was applied to show the data distribution both in case of categorical and numerical variables.

3. Correlation Analysis

Correlation analysis was performed to find the relationship between numerical variables. Numerical variables included age, BMI, and average glucose level, which were studied to identify the presence of multicollinearity.

4. Data Balancing (Upsampling)

To reduce class imbalance, we use upsampling so that each class contained 3,192 observations, ensuring unbiased model training.

5. Logistic Regression Analysis

As the primary statistical approach, logistic regression analysis was used to estimate the relationship between several independent variables and a binary variable (stroke). It allows revealing the impact of several predictor variables on an events probability occurrence.

6. Odds Ratio Analysis

In this study, coefficients obtained using logistic regression were converted into odds ratios to find the likelihood ratio. In other words, the odds ratio is a coefficient showing the likelihood ratio of event occurrence among subjects under certain conditions.

7. Prediction and Classification

For the prediction of stroke among individuals, the logistic regression model was applied. Using the predictions, individuals were categorized into either stroke or non-stroke groups.

8. Goodness-of-Fit (Hosmer-Lemeshow Test)

To prove the model was accurate, we used Hosemer- Lemeshow Test. We get a p-value greater than 0.05, then the model is said to be accurate (good fit).

1.5 SOFTWARE & PACKAGES USED

The data analysis was carried out using the R programming language, which is widely used for statistical computation and data manipulation. Functions such as prop.table() were used for proportion calculations, while data cleaning and filtering were performed using the dplyr package. Graphical representations were created using base R functions like histogram(), barplot(), and boxplot(), along with the ggplot2 package for improved visualization. Logistic regression modeling was performed using the glm function. Multicollinearity among variables was assessed using the car package, and the Hosmer–Lemeshow goodness-of-fit test was conducted using the ResourceSelection package. Model evaluation was carried out using the caret package, which was also used to handle class imbalance through upsampling. Additionally, the ROC curve and AUC values were computed using the pROC package.

CHAPTER 2

EXPLORATORY DATA ANALYSIS

2.1 DATA SOURCE & DESCRIPTION

The dataset used in this project is the publicly available Stroke Prediction Dataset sourced from Kaggle (originally prepared by McKinsey & Company). It contains medical records from 5,110 individuals and is designed for binary classification tasks: predicting whether a patient has had a stroke or not.

2.1.1 Dataset overview

- **Total Records:** 5,110 observations
- **Total Attributes:** 12 (including the target variable)
- **Target Variable:** stroke (1 = stroke occurred, 0 = no stroke)
- **Stroke Cases:** 249 (4.9% of total: minority class)
- **Non-Stroke Cases:** 4,861 (95.1%: majority class)

2.1.2 Data Description

- This contains healthcare-related information of individuals, including demographic, medical, and lifestyle attributes.
- The variables include both categorical and continuous features:
    - **Demographic variables:** Gender, Age, Residence type, Ever married
    - **Medical variables:** Hypertension, Heart disease
    - **Lifestyle variables:** Smoking status, Work type, BMI, Avg glucose level
    - **Target variable:** Stroke

- The dataset is imbalanced, as only a small proportion of individuals have experienced a stroke.
- Approximately 5% of the observations belong to the stroke class, while the remaining 95% do not.

2.1.3 Variables Description

The dataset contains the following features:

Variable	Type	Description
Id	Numeric	Unique ID for each patient
Gender	Categorical	Male, Female, Other
Age	Continuous	Age in years
Hypertension	Binary	0 = No, 1 = Yes
Heart disease	Binary	0 = No, 1 = Yes
Ever married	Categorical	Yes or No
Work type	Categorical	Private, Self-employed, Govt-job, Children, Never-worked
Residence type	Categorical	Urban or Rural
Avg glucose level	Continuous	Average blood glucose (mg/dL)
BMI	Continuous	Body Mass Index (kg/m^2)
Smoking status	Categorical	formerly smoked, never smoked, smokes, Unknown
Stroke	Binary	0 = No stroke, 1 = Stroke

Table 2.1-Variables Description

2.2 DESCRIPTIVE STATISTICS

The table below shows the key statistical measures for the continuous variables in the dataset. These measures help to identify the typical values for our participants and the range of health conditions present in the study.

Variables	Min	1st Q	Median	Mean	3rd Q	Max
Age	18.00	36.00	51.00	50.26	64.00	82.00
Avg glucose level	55.12	77.47	92.47	108.53	116.11	271.74
BMI	11.30	25.60	28.90	30.24	33.80	59.70

Table 2.2- Summary of Continuous Variables

Descriptive statistics were used to summarize the numerical variables in the dataset, including age, body mass index (BMI), and average glucose level. Measures such as mean, median, standard deviation, minimum, and maximum were computed to understand the central tendency and variability of the data.

The average age of individuals in the dataset indicates a wide age distribution, reflecting both younger and older populations. The BMI values show moderate variability, suggesting differences in body composition among individuals. Similarly, the average glucose level exhibits a noticeable spread, indicating variation in metabolic health across the population.

The comparison between mean and median values suggests the presence of slight skewness in some variables, particularly glucose levels, where a few high values may influence the mean.

2.3 CATEGORICAL VARIABLE ANALYSIS

2.3.1 Distribution of the Target Variable (Stroke)

The Target variable **Stroke,** a binary categorical variable which is represented by stroke (1) & no stroke (0). The dataset shows a severe imbalance.

Case	Count(n)	Percentage (%)
No Stroke (0)	4,861	95.1%
Stroke (1)	249	4.9%

Table 2.4 – Distribution of Stroke

2.2.3 Summary of Categorical Variables

Categorical variables represent the types of data which may be divided into groups. Unlike numerical data, these variables do not have a mathematical value but instead describe a quality or characteristic.

Category	Grouping / Levels	Count (n)	Percentage (%)
Gender	Female	2,994	58.6%
	Male	2,115	41.4%
	Other	1	0.02%
Ever Married	Yes (1)	3,353	65.6%
	No (0)	1,757	34.4%
Residence Type	Urban	2,596	50.8%
	Rural	2,514	49.2%
Work Type	Private	2,792	54.6%
	Self-employed	807	15.8%
	Govt job	651	12.8%
	Children	855	16.8%

	Never worked	5	0.09%
Smoking Status	Never Smoked	2,608	51.0%
	Unknown	862	16.9%
	Formerly Smoked	860	16.9%
	Smokes	780	15.2%
Hypertension	Yes (1)	498	9.7%
	No (0)	4,612	90.3%
Heart Disease	Yes (1)	276	5.4%
	No (0)	4,834	94.6%

Table 2.3- Summary of Categorical Variables

The distribution of categorical variables was examined using frequency tables. The dataset contains a larger proportion of non-stroke cases compared to stroke cases, indicating an imbalance in the target variable.

Gender distribution shows a fairly balanced representation of male and female individuals, with no significant dominance of either category.

In terms of hypertension, only a smaller proportion of individuals are identified as hypertensive; however, this group is clinically important due to its association with stroke risk.

Similarly, the presence of heart disease is less frequent in the dataset but remains a critical factor for analysis.

Smoking status was categorized into groups such as ‘never smoked’, ‘formerly smoked’, and ‘currently smoking’. The majority of individuals belong to the ‘never smoked’ category, followed by smaller proportions of former and current smokers

2.4 CORRELATION ANALYSIS

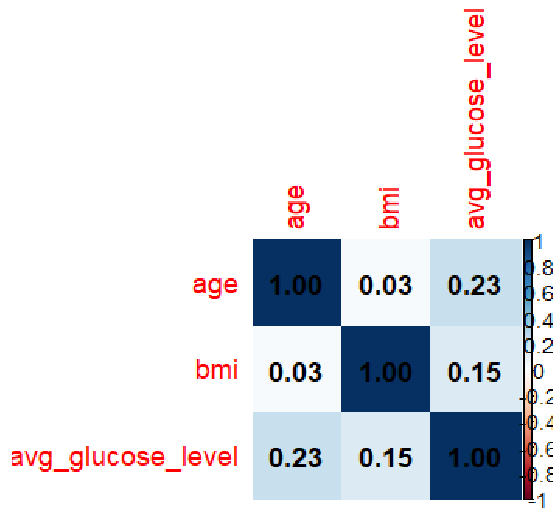


Figure 2.1 Correlation Matrix

The correlation matrix shows the relationship between the numerical variables. To quantify the linear relationships between the continuous variables, a Pearson correlation analysis was conducted. The resulting matrix reveals several key insights such as:

Weak Inter-variable Dependency: The most notable observation is the lack of strong correlation between the predictors. The highest correlation exists between Age and Average Glucose Level (0.23), which is logically sound as metabolic regulation often shifts with advancing age. However, a value of 0.23 is still considered a weak positive correlation, indicating that these variables provide distinct, non-redundant information.

Independence of BMI: Interestingly, BMI shows almost no correlation with age (0.03) and only a very slight positive relationship with Glucose (0.15). This suggests that in this dataset, body mass index is an independent physical metric that does not necessarily scale linearly with the other health indicators.

Absence of Multicollinearity: From a modeling perspective, the fact that all coefficients are well below the 0.7 or 0.8 threshold is highly positive. This confirms that the dataset does not suffer from multicollinearity, meaning that each variable can be included in the logistic regression model without risk of distorting the coefficient estimates or inflating the variance.

Overall, all variables show weak correlations with each other, indicating the absence of strong linear relationships. This suggests that multicollinearity is not a significant issue in the dataset.

CHAPTER 3

METHODOLOGY

3.1 DATA PREPARATION

Data Preparation constitute the most critical phase of analysis. It is in this stage that the raw, unbalanced data is transformed into a high-quality dataset.

3.1.1 Handling Missing Values

An initial inspection of the dataset revealed that 201 observations (approximately 3.9% of the total) contained missing values for the BMI variable. Since the proportion of missing data is below the commonly accepted threshold of 5%, and there is no systematic pattern to the missingness, a median imputation strategy was adopted. Median imputation is preferred over mean imputation in the presence of skewed distributions, as it is more robust to extreme values. The median BMI computed from the non-missing observations was 28.1 kg/m², and this value was substituted for all 201 missing records.

3.1.2 Filtering by Age, BMI, Work type and Gender

Records belonging to individuals aged below 18 years were excluded from the analysis, as stroke risk patterns in this populations due to the other factors differ substantially from those in adults and would introduce heterogeneity that is beyond the scope of this study. Additionally, records with BMI values below 10 or above 60 kg/m² were excluded as these represent physiologically improbable values and were treated as data entry errors. Further, we removed participants categorized under 'Never Worked' and ‘Other’ category from gender. This was done because these groups represented an extremely small fraction of the entire data, and their inclusion could introduce statistical noise without providing enough evidence for the model to learn meaningful patterns.

3.1.3 Variable Encoding

Categorical variables including gender, Work type, Residence type, Smoking status, and ever married were converted to factor variables in R using 'as.factor()'. The stroke variable was encoded as a binary factor with levels '0' (no stroke) and '1' (stroke) to serve as the dependent variable in the logistic regression model. It is also converted to factor. The id variable was dropped entirely as it carries no predictive information for the model. After applying all these filters, the final dataset comprised of 4,236 observations.

3.1.5 Data Splitting (Train-Test Split)

Once the data was cleaned, encoded, and standardized, it was partitioned into two independent subsets: a **Training Set** (70%) and a **Testing Set** (30%).

The cleaned dataset (N = 4,236) was partitioned into a Training Set (70%, n = 2,966) and a Testing Set (30%, n = 1,270) using stratified random sampling. Stratification was employed to ensure that the distribution of stroke cases remained consistent across both subsets. The training set was reserved for model development and coefficient estimation, while the test set remained "unseen" to provide an unbiased evaluation of the model's predictive performance on new, real-world data.

3.1.6 Handling Imbalanced Data (Upsampling)

To reduce bias toward the majority 'no-stroke' class, random upsampling was applied only to the training dataset. In this process, the number of stroke cases (minority class) was increased until it matched the number of non-stroke cases by taking duplicates of the data. As a result, the training dataset became balanced, with 3,192 observations in each class, giving a total of 6,384 observations. The test dataset was kept unchanged and remained imbalanced. This approach ensures that the model is evaluated under realistic conditions, allowing performance measures such as sensitivity and precision to better reflect its practical usefulness. After performing these preprocessing steps, the dataset now became clean, balanced, and suitable for further analysis and modeling.

3.2 STATISTICAL FRAMEWORK: LOGISTIC REGRESSION

3.2.1 Model Specification

Logistic regression is a generalized linear model designed for binary outcome variables. Given a set of predictor variables $X_1, X_2, \ldots, X_k$, the logistic regression model estimates the probability that the binary outcome Y equals 1 (in this case, the probability of stroke) using the logistic (sigmoid) function:

$$P(Y = 1 \mid X) = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k)})$$

where β_0 is the intercept and $\beta_1, \ldots, \beta_k$ are the coefficients associated with each predictor. The log-odds (logit) transformation linearizes this relationship:

$$\log[P/(1-P)] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k$$

The model parameters are estimated using maximum likelihood estimation (MLE), which finds the coefficient values that maximize the likelihood of observing the given data. The exponential of each coefficient, e^{β_j}, gives the odds ratio (OR) for predictor X_j, interpreted as the multiplicative change in the odds of stroke per unit increase in X_j, holding all other predictors constant. An odds ratio greater than 1 indicates increased risk, while a value less than 1 indicates decreased risk.

3.2.2 Multicollinearity Check (VIF)

Multicollinearity among the predictor variables was assessed using the Variance Inflation Factor (VIF). High multicollinearity can affect the stability and interpretation of regression coefficients. VIF values close to 1, indicates that there is no significant multicollinearity among the variables.

3.3 Model Evaluation Metrics

The performance of the logistic regression model was evaluated using the test dataset with the help of the following metrics.

- Confusion Matrix: A 2×2 matrix tabulating true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).
- Accuracy: (TP + TN) / (TP + TN + FP + FN) — the proportion of all predictions that were correct.
- Sensitivity (Recall): TP / (TP + FN) — the proportion of actual stroke cases correctly identified.
- Specificity: TN / (TN + FP) — the proportion of actual non-stroke cases correctly identified.
- Precision: TP / (TP + FP) — the proportion of predicted stroke cases that were true positives.
- F1-Score: 2 × (Precision × Sensitivity) / (Precision + Sensitivity) — the harmonic mean balancing precision and recall.
- AUC-ROC: The Area Under the Receiver Operating Characteristic Curve, measuring the model's ability to discriminate between stroke and non-stroke cases across all classification thresholds. Computed using the pROC package.

3.4 Model Calibration and Validation

While standard performance metrics assess the model's discriminative power, Model Calibration evaluates the reliability of the predicted probabilities. In clinical settings, calibration is paramount because it ensures that the estimated risk corresponds accurately to the observed frequency of the event.

3.4.1 The Hosmer-Lemeshow Goodness-of-Fit Test

The **Hosmer-Lemeshow (H-L) test** is utilized to determine how well the model fits the data. The test works by partitioning the observations into deciles based on their predicted probabilities and comparing the "observed" stroke cases to the "expected" stroke cases in each group.

The Null Hypothesis (H_0)**:** The model's predicted probabilities match the observed stroke frequencies (Perfect fit).

Interpretation**:** Unlike many statistical tests where a low p-value is desired, for the H-L test, a non-significant p-value ($p > 0.05$) is the goal. A high p-value indicates that there is no significant difference between the model's predictions and reality, confirming that the logistic regression model is well-calibrated and mathematically sound.

CHAPTER 4

RESULTS & DISCUSSION

4.1 LOGISTIC REGRESSION RESULTS

4.1.1 Model Coefficients

Variable	Estimate	Std. Error	z value	p-value
(Intercept)	-5.4779	0.2524	-21.706	<0.001
gender (Male)	0.0496	0.0630	0.787	0.4313
age	0.0752	0.0024	31.573	<0.001
hypertension	0.4462	0.0813	5.490	<0.001
heart disease	0.3135	0.1008	3.108	0.0019
ever married	0.0457	0.0989	0.462	0.6441
work type (Private)	0.3424	0.0897	3.816	<0.001
work type (Self-employed)	0.0721	0.1036	0.696	0.4866
residence (Urban)	0.1332	0.0614	2.171	0.0299
avg glucose level	0.0045	0.0006	7.781	<0.001
BMI	-0.0012	0.0051	-0.244	0.8072
smoking (never smoked)	-0.1310	0.0802	-1.635	0.1021
smoking (smokes)	0.3390	0.0950	3.570	<0.001
smoking (Unknown)	-0.0198	0.0945	-0.210	0.8337

Table 4.1 Model Coefficients Table

The logistic regression model was fitted to identify the relationship between predictor variables and stroke occurrence. The estimated coefficients, along with their standard errors, z-values, and p-values, are presented in the table above.

Variables such as age, hypertension, heart disease, work type (Private), residence type (Urban), average glucose level, and smoking status (smokes) were found to be statistically significant ($p < 0.05$), indicating a strong association with stroke occurrence. On the other hand, variables such as gender, BMI, marital status, and some smoking categories were not found to be statistically significant, suggesting a weaker or negligible influence on stroke in this dataset.

4.1.2 Odds Ratio Interpretation

Predictor	**Odds Ratio**	**95% CI (Lower, Upper)**	**Interpretation**
Age	1.078	[1.073, 1.083]	Strongest continuous predictor. Risk increases by 7.8% for every year of age.
Hypertension	1.562	[1.333, 1.833]	Significant risk factor. Hypertensive patients are 56.2% more likely to suffer a stroke.
Heart Disease	1.368	[1.125, 1.670]	Significant risk factor. Presence of heart disease increases stroke odds by 36.8%.

Predictor	Odds Ratio	95% CI (Lower, Upper)	Interpretation
Avg Glucose Level	1.005	[1.003, 1.006]	Statistically significant but with a small marginal effect per unit increase.
Smoking (Smokes)	1.404	[1.165, 1.691]	High risk. Active smokers have 40.4% higher odds compared to former smokers.
Work Type (Private)	1.408	[1.182, 1.680]	Significant association; may be tied to lifestyle or stress factors.
BMI	0.999	[0.989, 1.009]	Insignificant. The CI includes 1.0, suggesting BMI is not an independent predictor here.
Gender (Male)	1.051	[0.929, 1.189]	Insignificant. No statistical difference in stroke risk between genders.

Table 4.2 Odds Ratio Table

The odds ratio table represents the effect of each predictor variable on the likelihood of stroke. An odds ratio greater than 1 indicates an increased risk of stroke, while a value less than 1 indicates a protective or reduced effect. From this table it is clear that age is the most significant predictor, with an odds ratio greater than 1, indicating that the risk of stroke increases with age. Similarly, hypertension and heart disease show strong positive associations with stroke occurrence. Average glucose level and smoking status (smokes) also show increased odds, suggesting their importance as lifestyle and metabolic risk factors. Variables such as BMI, gender, and some smoking categories show odds ratios close to 1, indicating weak or no significant association with stroke in this dataset.

Overall, the odds ratio analysis confirms that age, hypertension, heart disease, glucose level, and smoking status are the major predictors of stroke in this study.

4.2 MODEL PERFORMANCE EVALUATION

The predictive performance of the logistic regression model was evaluated using the test dataset. Various evaluation metrics were used to assess how accurately the model classifies stroke and non-stroke cases.

4.2.1 Confusion Matrix

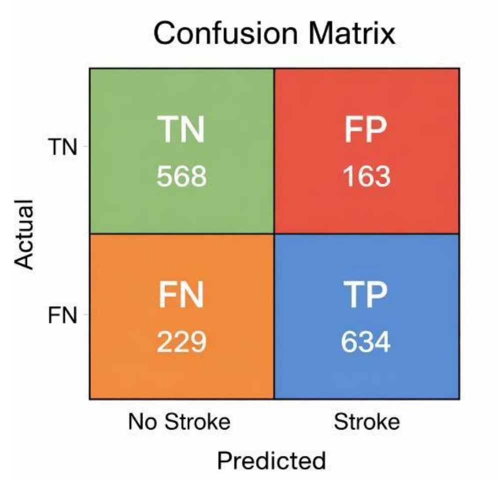


Table 4.3 Confusion Matrix

The confusion matrix summarizes the classification performance of the model by comparing predicted and actual outcomes. The model correctly identified 634 stroke cases (true positives) and 568 non-stroke cases (true negatives). However, 229 stroke cases were misclassified as non-stroke (false negatives), and 163 non-stroke cases were incorrectly classified as stroke (false positives)

4.2.1 Performance Matrix

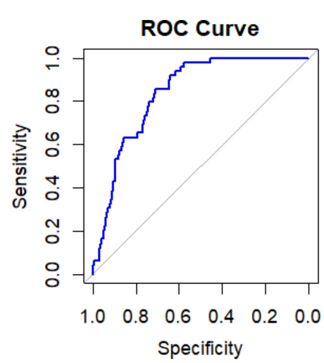
Metric	Value
Accuracy	75.4%
Sensitivity	73.5%
Specificity	77.7%
Precision	79.5%
F1 Score	76.4%
AUC	0.8395

Table 4.4 Performance Matrix

The model achieved an accuracy of 75.4%, indicating that a majority of the observations were correctly classified. The sensitivity value shows that the model is capable of identifying stroke cases effectively, while the specificity indicates good performance in identifying non-stroke cases.

The precision and F1-score demonstrate that the model maintains a good balance between detecting true stroke cases and minimizing incorrect predictions. The AUC value of 0.8395 further confirms that the model has strong classification ability.

4.2.2 ROC Curve and AUC



AUC value = 0.8395

Figure 4.1 ROC Curve

The Receiver Operating Characteristic (ROC) curve was used to evaluate the performance of the logistic regression model across different classification thresholds. The curve represents the trade-off between sensitivity and specificity. The Y-Axis (Sensitivity) represents the "True Positive Rate." It shows how well the model identifies patients who actually had a stroke. The X-Axis (Specificity) shows specificity from 1.0 to 0.0. As the curve moves toward the left, the model is successfully identifying stroke patients without misclassifying too many healthy people as "at risk."

The Diagonal Line, a thin gray line represents a "random guesser." Since the blue line is significantly above this diagonal, it proves that Logistic Regression model has learned strong patterns from the data.

The Area Under the Curve (AUC) was found to be 0.8395, which indicates good predictive performance. A value closer to 1 suggests a strong ability of the model to distinguish between stroke and non-stroke cases.

4.3 HOSMER–LEMESHOW GOODNESS-OF-FIT TEST

The Hosmer–Lemeshow goodness-of-fit test was performed to assess how well the logistic regression model fits the data. This test compares the observed and predicted probabilities of stroke across different groups. The test resulted in a Chi-square value of 6.0459 with a p-value of 0.6421.

If $p > 0.05$, then there is no significant difference between observed and predicted values, indicating that the model fits the data well and If $p < 0.05$, indicating that the model does not fit the data adequately. Here p value is greater than 0.05, therefore the model fits the data well.

4.4 MULTICOLLINEARITY (VIF)

Multicollinearity among predictor variables was assessed using the Variance Inflation Factor (VIF). High multicollinearity can affect the stability and interpretability of the regression coefficients.

Variable	GVIF^(1/(2*Df))
Age	1.11
Hypertension	1.03
Heart disease	1.05
Avg Glucose level	1.06
BMI	1.04
Gender	1.02
Work type	1.03
Smoking status	1.02
Residence type	1.01
Ever married	1.03

Table 4.5 VIF

A VIF value greater than 5 (or 10) typically indicates the presence of multicollinearity. Here the VIF values for all variables were found to be close to 1, indicating that there is no significant multicollinearity among the predictor variables.

CHAPTER 5

CONCLUSION & FUTURE SCOPE

5.1 CONCLUSION

This project focused on developing a predictive model for stroke using logistic regression, based on demographic, lifestyle, and clinical factors. The analysis was carried out on a properly cleaned and preprocessed dataset, which helped ensure the accuracy and reliability of the results.

The results clearly show that age is the most important factor influencing stroke, with the risk increasing as age advances. Medical conditions such as hypertension and heart disease were also found to have a strong relationship with stroke occurrence. In addition, higher average glucose levels were associated with an increased risk of stroke.

Among lifestyle factors, smoking was found to contribute to stroke risk, while variables like gender and BMI did not show a significant impact in this particular dataset.

The performance of the model was evaluated using standard metrics. The model achieved an accuracy of 75.4%, along with balanced sensitivity and specificity. The AUC value of 0.8395 indicates that the model has a good ability to distinguish between stroke and non-stroke cases.

Further validation using the Hosmer–Lemeshow test showed that the model fits the data well. The multicollinearity check using VIF confirmed that there is no strong correlation among the predictor variables, ensuring the stability of the model.

5.2 LIMITATIONS OF THE STUDY

Despite the useful findings, the study has certain limitations. The dataset used may not fully represent the entire population, which can affect the generalizability of the results. Additionally, some potentially important factors such as genetic information, dietary habits, and physical activity levels were not included in the analysis.

5.3 FUTURE SCOPE

Future research can focus on improving prediction accuracy by applying advanced machine learning techniques such as Random Forest, Support Vector Machines, and Neural Networks.

Incorporating additional health-related variables and real-time medical data could further enhance the predictive capability of the model.

The model can also be extended into a practical application, such as a web-based or mobile-based tool, which can assist healthcare professionals in identifying high-risk individuals at an early stage.

Furthermore, validation of the model using larger and more diverse datasets would improve its reliability and applicability in real-world clinical settings.

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